THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL UNIT TEST SET 1

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The angle between the bisectors of the two angles of a linear pair is: 1

(a) 180° **(b)** 90° **(c)** 45° **(d)** 60°

2. The smallest 3-digit number divisible by 9 is: 1

(a) 107 **(b)** 108 **(c)** 109 **(d)** 117

3. The number of triangles with whole-number sides (in cm) and perimeter 9 cm is: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

4. $\frac{3}{4} \div (\frac{3}{2} \div \frac{1}{2})$ equals: 1

(a) $\frac{1}{8}$ **(b)** 1 **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{2}$

5. Two supplementary angles are in the ratio $1 : 2$. The larger angle is: 1

(a) 120° **(b)** 140° **(c)** 130° **(d)** 110°

6. The largest 4-digit number with all digits different that is divisible by 9 is: 1

(a) 9872 **(b)** 9874 **(c)** 9873 **(d)** 9876

7. The exterior angle at the base of an isosceles triangle is 110°. The apex angle is: 1

(a) 40° **(b)** 60° **(c)** 50° **(d)** 30°

8. $\frac{2}{5}$ of a number is 18. Three times the number is: 1

(a) $67\frac{1}{2}$ **(b)** 270 **(c)** 135 **(d)** $135\frac{1}{2}$

9. The supplement of the complement of 20° is: 1

(a) 130° **(b)** 100° **(c)** 120° **(d)** 110°

10. Kaprekar's routine on 2111 reaches 6174 after how many steps? 1

(a) 4 **(b)** 6 **(c)** 5 **(d)** 7

11. In $\triangle ABC$, $\angle A = \angle B + \angle C$. Then $\angle A$ is: 1

(a) 90° **(b)** 60° **(c)** 45° **(d)** 120°

12. A pipe fills $\frac{2}{5}$ of a tank in 12 minutes. The time to fill the whole tank (minutes) is: 1

(a) 60 **(b)** 15 **(c)** 30 **(d)** $30\frac{1}{2}$

13. The reflex angle that goes with an angle of 70° (so that together they make a complete turn) is: 1

(a) 290° **(b)** 280° **(c)** 310° **(d)** 300°

14. The Collatz sequence from 12 needs how many steps to reach 1? 1

(a) 8 **(b)** 10 **(c)** 9 **(d)** 11

15. The angles of a triangle are $(2x)^\circ$, $(3x+5)^\circ$ and $(4x-50)^\circ$. The largest angle is: 1

(a) 70° **(b)** 100° **(c)** 90° **(d)** 80°

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the angle whose supplement is 3 times its complement. 2

17. A two-digit number added to its reverse gives 110 and subtracted from it gives 36. Find the number. 2

18. In $\triangle ABC$ and $\triangle PQR$ it is known that $AB = PQ = 5$ cm, $BC = QR = 6$ cm and $\angle A = \angle P = 50^\circ$. Are the triangles necessarily congruent? Give the rule or the reason. 2

19. Simplify $\left(\frac{5}{6}+\frac{1}{3}\right) \div \left(\frac{5}{6}-\frac{1}{3}\right) \times \frac{1}{2}$. 2

20. An angle is 6° more than 3 times its complement. Find the angle. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Angles on one side of a straight line at a point are 35°, 45°, 50° and a fourth angle. Find the fourth angle and name the kind of each angle. 3

22. The number $4 * 3 * 6$ has the same digit $*$ in both blanks and is divisible by 9. Find all possible values of $*$. 3

23. In $\triangle ABC$, $\angle A = 50^\circ$. The exterior angles at B and C are $(4x+60)^\circ$ and $(3x+100)^\circ$. Find x and the angles B and C. 3

24. Ravi alone can finish a job in 6 days and Sita alone in 3 days. What fraction of the job do they finish together in one day, and how many days do they take together? 3

OR

Two supplementary angles are in the ratio $1:4$. Find the complement of the smaller angle and the supplement of the larger angle. 3

25. Start with 1235 and apply Kaprekar's routine until 6174 appears. Show every step. 3

OR

The perimeter of an isosceles triangle is 40 cm. Each equal side is 2 cm longer than the base. Find the sides and show that such a triangle exists. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Three angles on one side of a straight line at a point are in the ratio $1:2:3$. Find the three angles and name the kind of each. 5

OR

Use reverse-and-add on 69 until a palindrome is formed. Show each step, state the number of steps and the final palindrome. 5

27. The exterior angles of a triangle (one at each vertex) are in the ratio $3:4:5$. Find the three exterior angles, the three interior angles and classify the triangle. 5

OR

A farmer's land is 240 acres. He sells $\frac{1}{4}$ of it, gives $\frac{1}{3}$ of the remainder to his son and plants $\frac{3}{4}$ of what is left after that with sugarcane. Find each area and the area left unplanted. 5

END OF QUESTION PAPER **Total: 50 Marks**

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